MongoDB Lab – Day 3 (Indexing & Sorting)

Schema

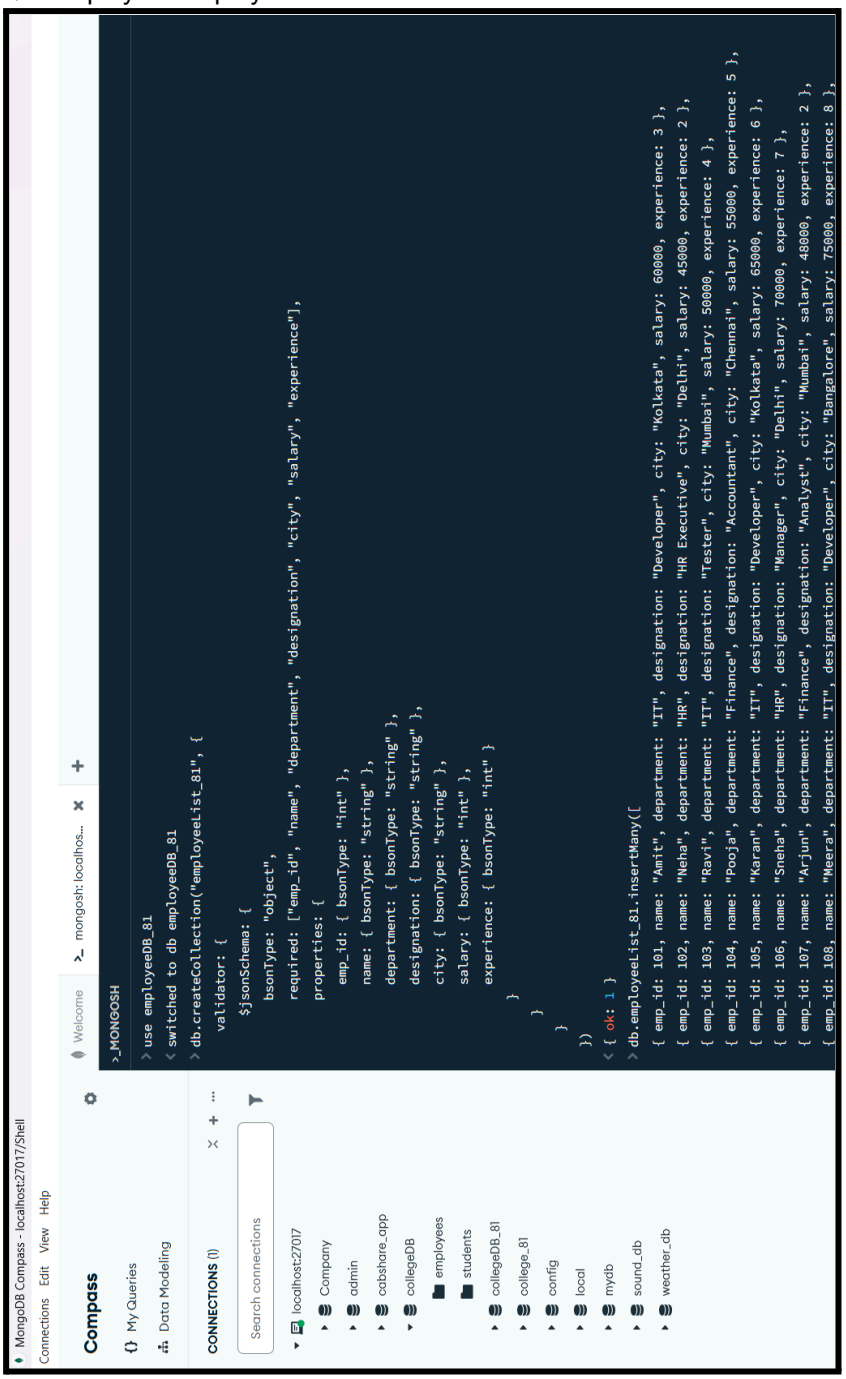
{ emp_id, name, department, designation, city, salary, experience }

Initial Data

Emp ID	Name	Department	Designation	City	Salary	Experience
101	Amit	IT	Developer	Kolkata	60000	3
102	Neha	HR	HR Executive	Delhi	45000	2
103	Ravi	IT	Tester	Mumbai	50000	4
104	Pooja	Finance	Accountant	Chennai	55000	5
105	Karan	IT	Developer	Kolkata	65000	6
106	Sneha	HR	Manager	Delhi	70000	7
107	Arjun	Finance	Analyst	Mumbai	48000	2
108	Meera	IT	Developer	Bangalore	75000	8
109	Rohit	IT	Team Lead	Pune	90000	10
110	Anjali	HR	Recruiter	Kolkata	42000	1

Lab Questions

Q1. Display all employee records.



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Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- localhost:27017
  - Company
  - admin
  - cabshare_app
  - collegeDB
    - employees
    - students
  - collegeDB_81
  - college_81
  - config
  - local
  - mydb
  - sound_db
  - weather_db

Welcome | mongosh: localhos...

>_MONGOSH

```
  { emp_id: 108, name: "Meera", department: "IT", designation: "Developer", city: "Bangalore", salary: 75000, experience: 8 },
  { emp_id: 109, name: "Rohit", department: "IT", designation: "Team Lead", city: "Pune", salary: 90000, experience: 10 },
  { emp_id: 110, name: "Anjali", department: "HR", designation: "Recruiter", city: "Kolkata", salary: 42000, experience: 1 }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('699b5897272a21068ee6f40e'),
      '1': ObjectId('699b5897272a21068ee6f40f'),
      '2': ObjectId('699b5897272a21068ee6f410'),
      '3': ObjectId('699b5897272a21068ee6f411'),
      '4': ObjectId('699b5897272a21068ee6f412'),
      '5': ObjectId('699b5897272a21068ee6f413'),
      '6': ObjectId('699b5897272a21068ee6f414'),
      '7': ObjectId('699b5897272a21068ee6f415'),
      '8': ObjectId('699b5897272a21068ee6f416'),
      '9': ObjectId('699b5897272a21068ee6f417')
    }
  }
> db.employeeList_81.find()
< {
    _id: ObjectId('699b5897272a21068ee6f40e'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
```

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Welcome | mongosh: localhost:27017

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```
    city: 'Chennai',
    salary: 55000,
    experience: 5
  }
  {
    _id: ObjectId('699b5897272a21068ee6f412'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  }
  {
    _id: ObjectId('699b5897272a21068ee6f413'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  }
  {
    _id: ObjectId('699b5897272a21068ee6f414'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
```

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```
    experience: 3
  }
  {
    _id: ObjectId('699b5897272a21068ee6f40f'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  }
  {
    _id: ObjectId('699b5897272a21068ee6f410'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  }
  {
    _id: ObjectId('699b5897272a21068ee6f411'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
```

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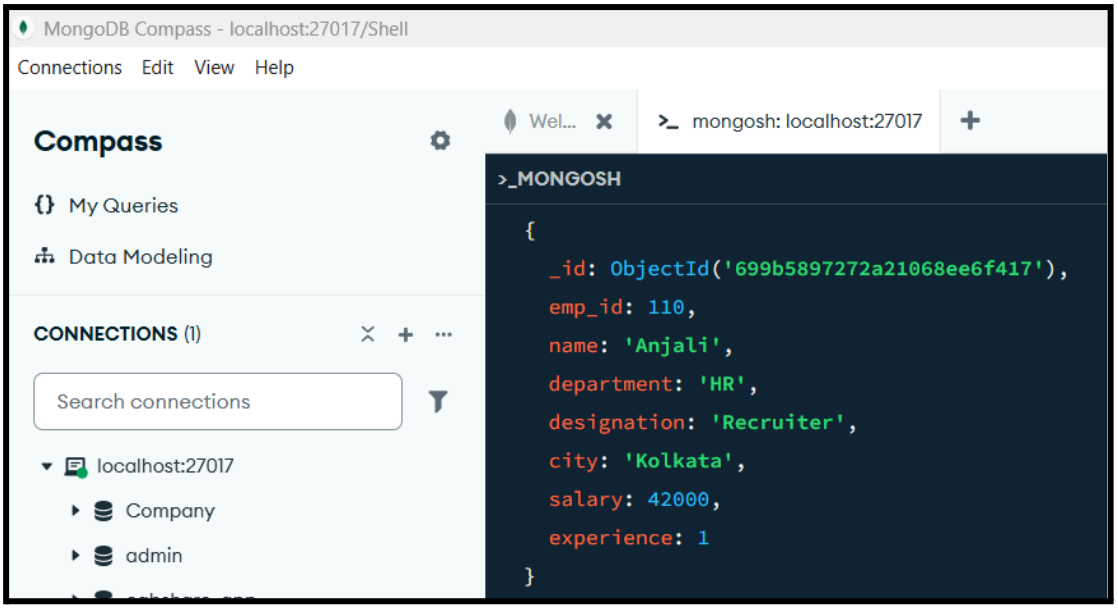
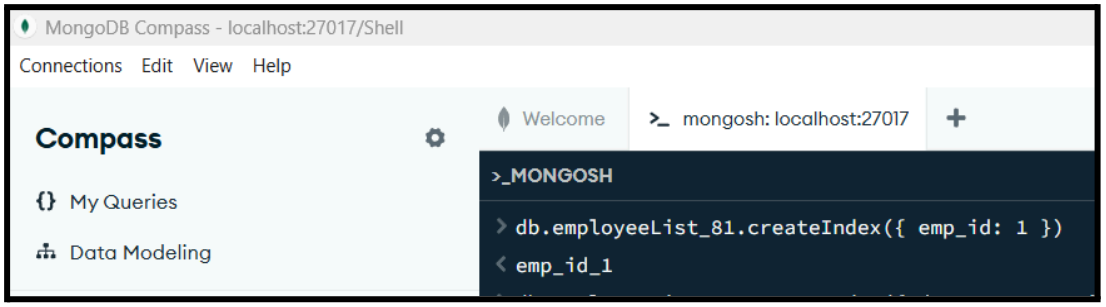
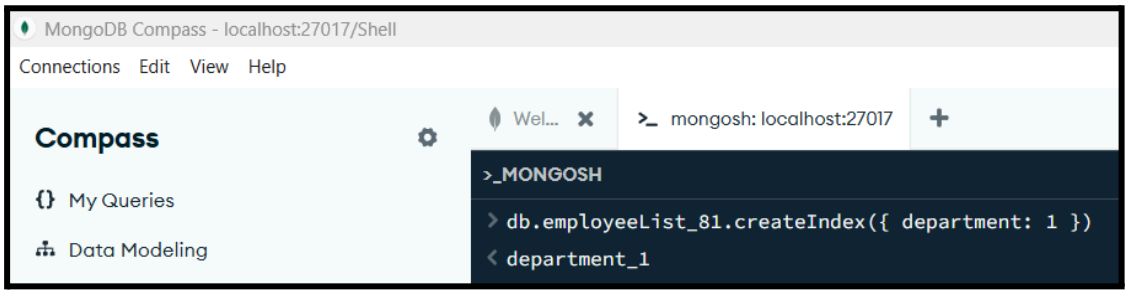
```
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  }
  {
    _id: ObjectId('699b5897272a21068ee6f415'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  }
  {
    _id: ObjectId('699b5897272a21068ee6f416'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
  {
    _id: ObjectId('699b5897272a21068ee6f417'),
    emp_id: 110,
```



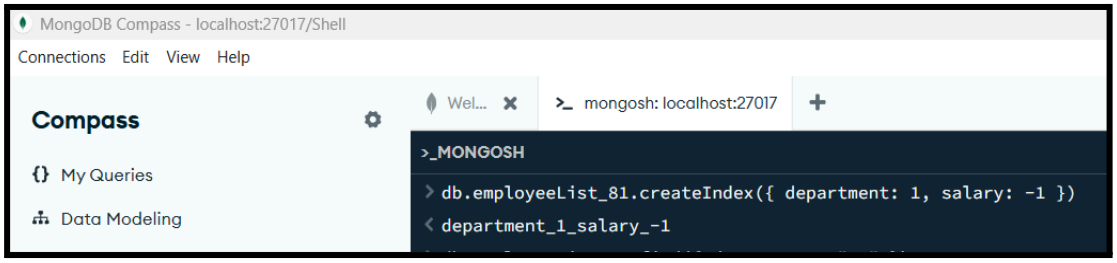
Q2. Create an index on emp_id field.



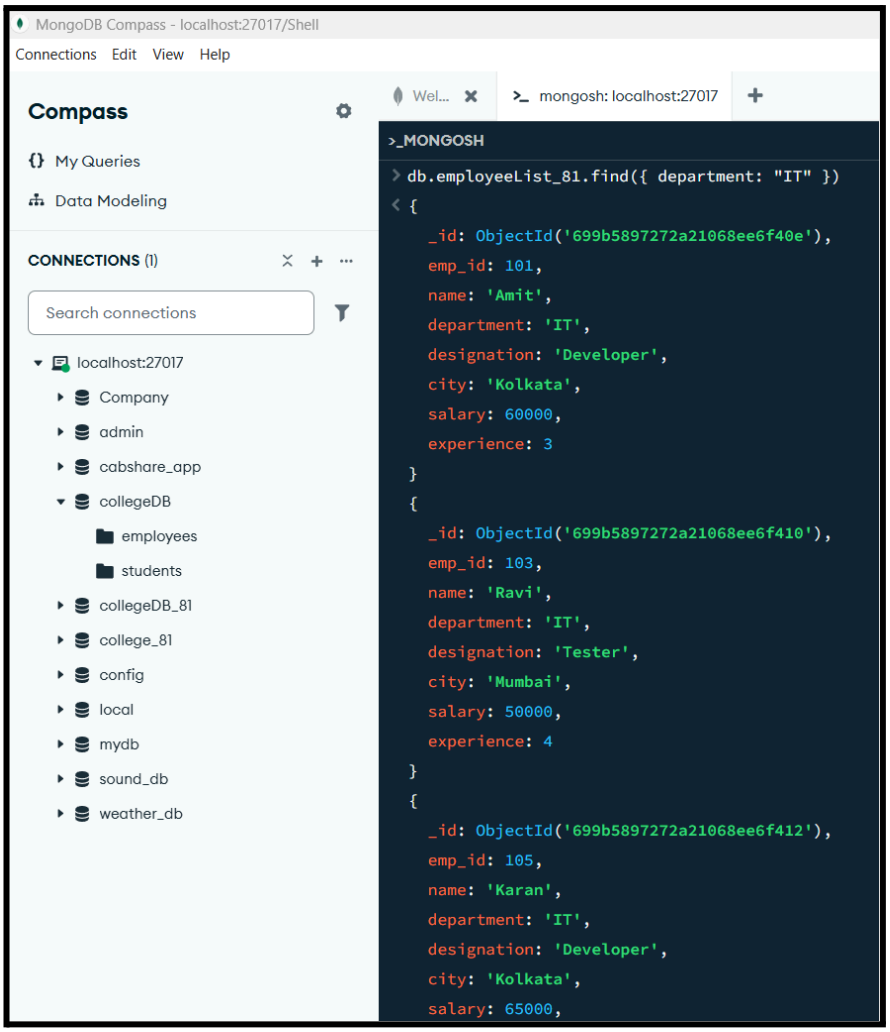
Q3. Create an index on department field.



Q4. Create a compound index on department and salary.



Q5. Find employees working in IT department.



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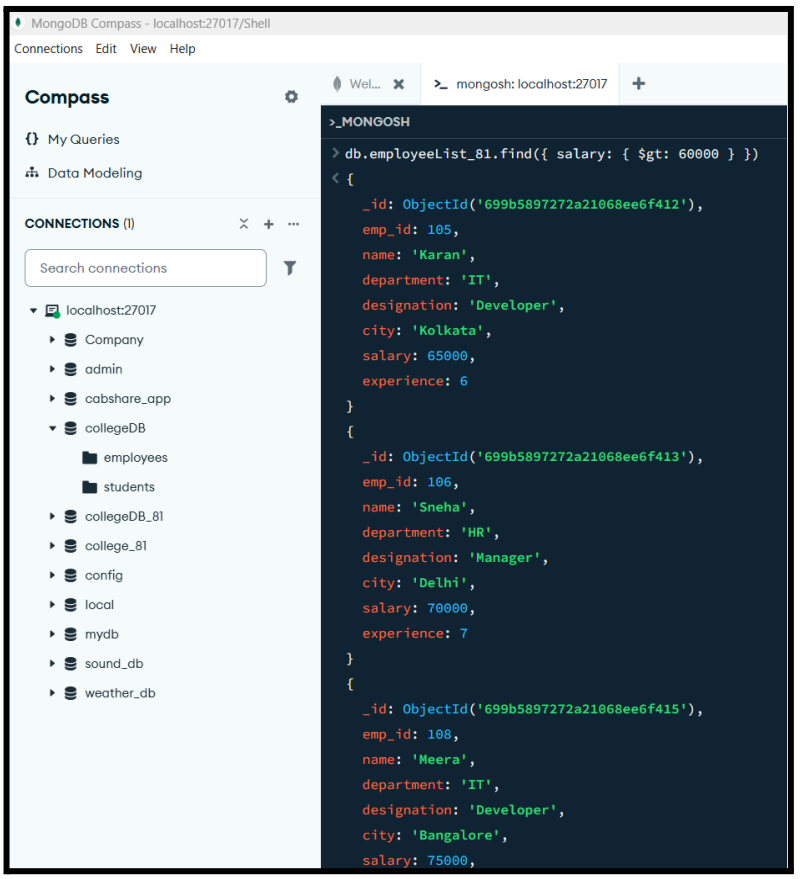
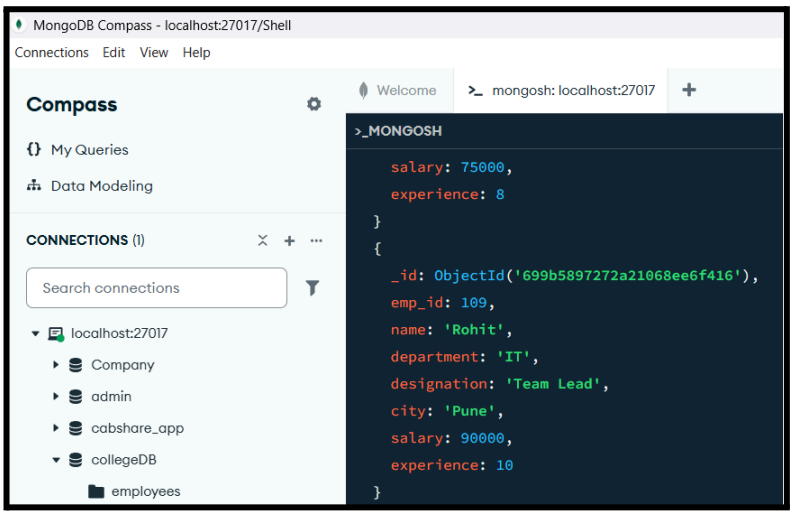
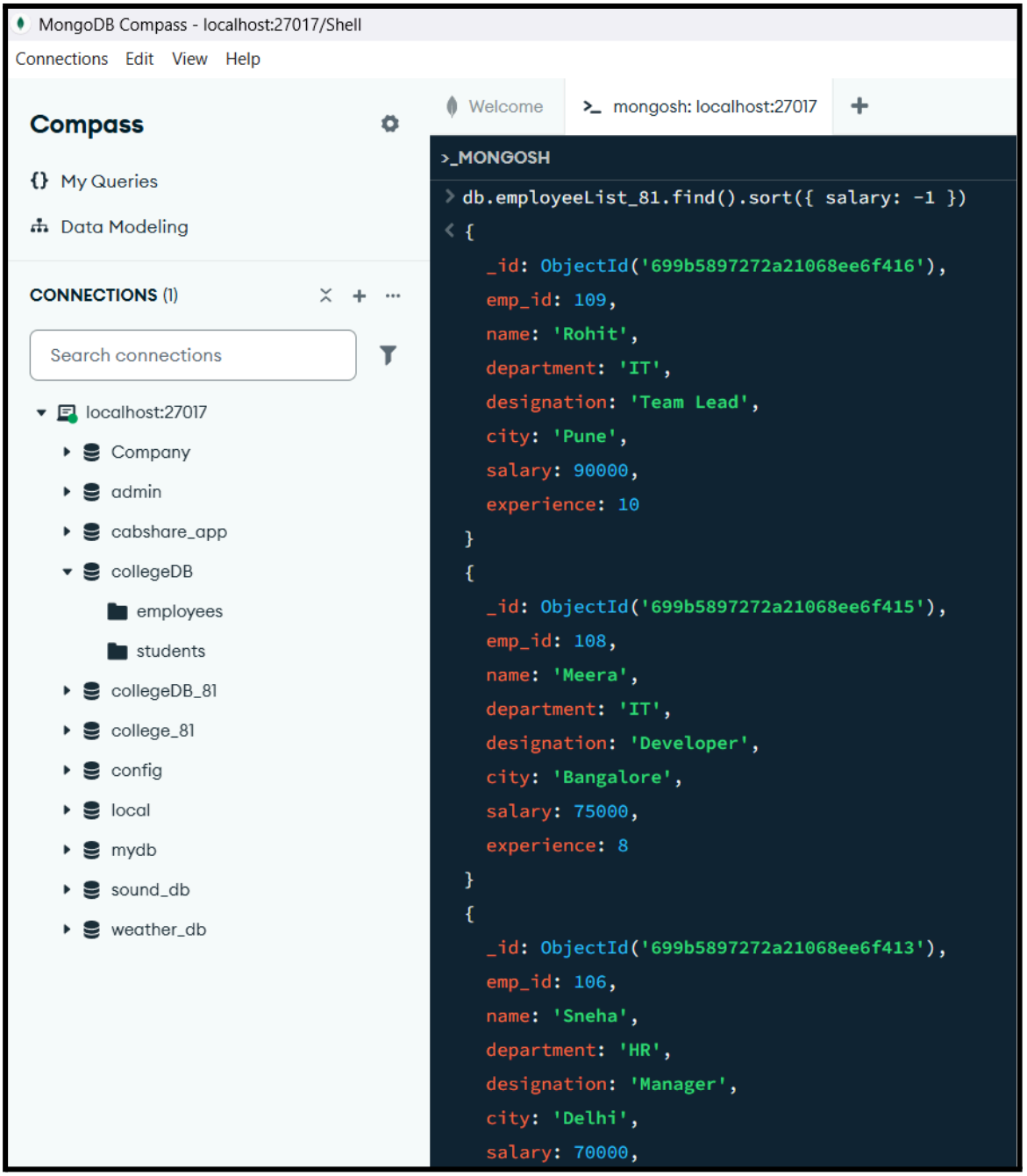
```
    salary: 65000,
    experience: 6
  }
  {
    _id: ObjectId('699b5897272a21068ee6f415'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  }
  {
    _id: ObjectId('699b5897272a21068ee6f416'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
```

Q6. Display employees whose salary is greater than 60,000.





Q7. Sort employees by salary in descending order.



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```
    salary: 70000,
    experience: 7
  }
  {
    _id: ObjectId('699b5897272a21068ee6f412'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  }
  {
    _id: ObjectId('699b5897272a21068ee6f40e'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  }
  {
    _id: ObjectId('699b5897272a21068ee6f411'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
```

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Wel... | mongosh: localhost:27017

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```
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  }
  {
    _id: ObjectId('699b5897272a21068ee6f410'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  }
  {
    _id: ObjectId('699b5897272a21068ee6f414'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  }
  {
    _id: ObjectId('699b5897272a21068ee6f40f'),
    emp_id: 102,
    name: 'Neha',
```

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My Queries

Data Modeling

CONNECTIONS (1)

Search connections

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- collegeDB_81
- college_81

Welcome | mongosh: localhost:27017

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```
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  }
  {
    _id: ObjectId('699b5897272a21068ee6f417'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
```

Q8. Display top 3 highest paid employees.

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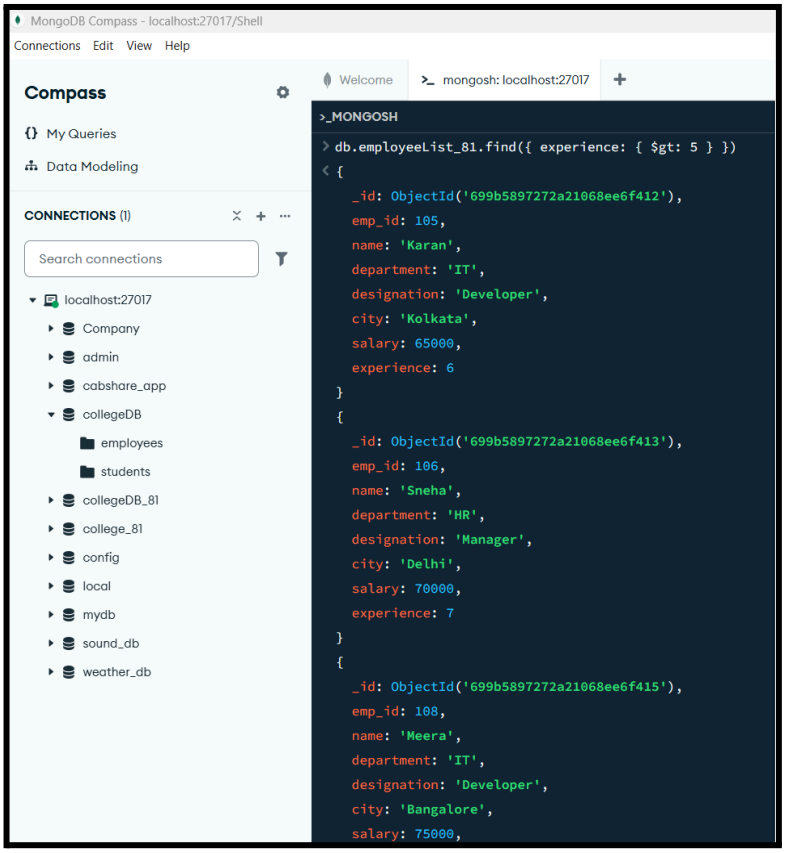
- weather_db

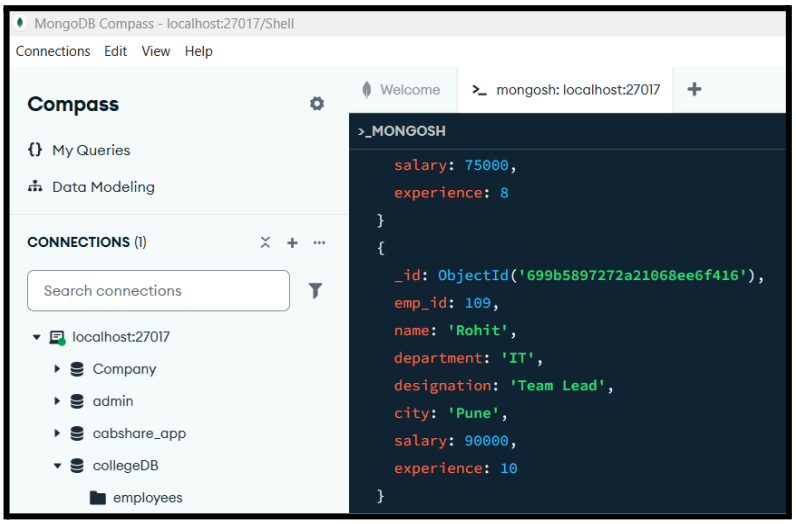
Welcome | mongosh: localhost:27017

>_MONGOSH

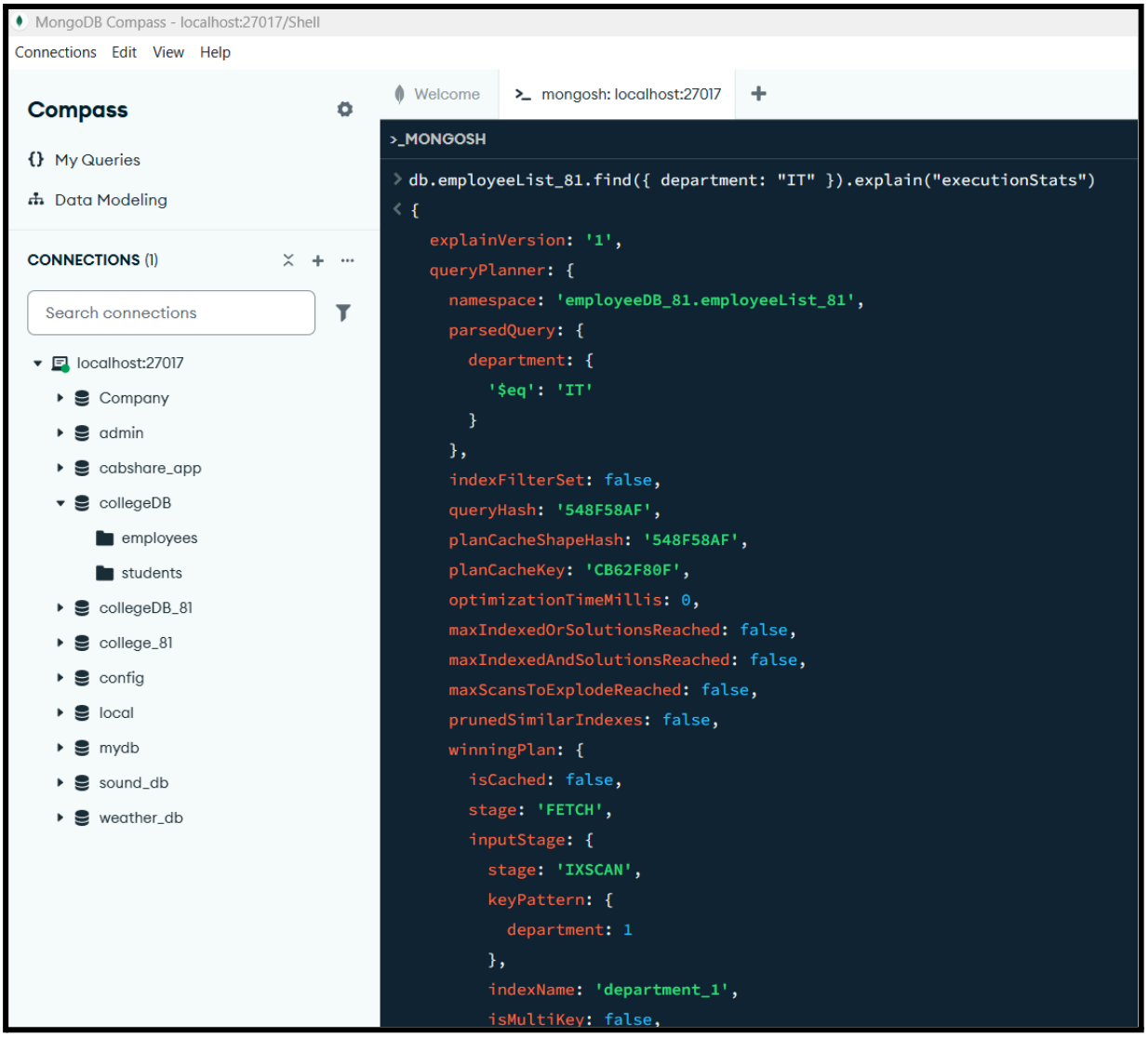
```
> db.employeeList_81.find().sort({ salary: -1 }).limit(3)
< {
    _id: ObjectId('699b5897272a21068ee6f416'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
  {
    _id: ObjectId('699b5897272a21068ee6f415'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  }
  {
    _id: ObjectId('699b5897272a21068ee6f413'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
```

Q9. Display employees having experience more than 5 years.





Q10. Analyze query performance using explain().



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Welcome | mongosh: localhost:27017

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```
        isMultiKey: false,
        multiKeyPaths: {
          department: []
        },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: {
          department: [
            '["IT", "IT"]'
          ]
        }
      }
    },
    rejectedPlans: [
      {
        isCached: false,
        stage: 'FETCH',
        inputStage: {
          stage: 'IXSCAN',
          keyPattern: {
            department: 1,
            salary: -1
          },
          indexName: 'department_1_salary_-1',
          isMultiKey: false,
          multiKeyPaths: {
```

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```
            multiKeyPaths: {
              department: [],
              salary: []
            },
            isUnique: false,
            isSparse: false,
            isPartial: false,
            indexVersion: 2,
            direction: 'forward',
            indexBounds: {
              department: [
                '["IT", "IT"]'
              ],
              salary: [
                '[MaxKey, MinKey]'
              ]
            }
          }
        }
      ]
    },
    executionStats: {
      executionSuccess: true,
      nReturned: 5,
      executionTimeMillis: 6,
      totalKeysExamined: 5,
      totalDocsExamined: 5,
      executionStages: {
        isCached: false,
```

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Welcome | >_ mongosh: localhost:27017

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```
        isCached: false,
        stage: 'FETCH',
        nReturned: 5,
        executionTimeMillisEstimate: 0,
        works: 7,
        advanced: 5,
        needTime: 0,
        needYield: 0,
        saveState: 0,
        restoreState: 0,
        isEOF: 1,
        docsExamined: 5,
        alreadyHasObj: 0,
        inputStage: {
          stage: 'IXSCAN',
          nReturned: 5,
          executionTimeMillisEstimate: 0,
          works: 6,
          advanced: 5,
          needTime: 0,
          needYield: 0,
          saveState: 0,
          restoreState: 0,
          isEOF: 1,
          keyPattern: {
            department: 1
          },
          indexName: 'department_1',
          isMultiKey: false,
```

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```
        isMultiKey: false,
        multiKeyPaths: {
          department: []
        },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: {
          department: [
            '["IT", "IT"]'
          ]
        },
        keysExamined: 5,
        seeks: 1,
        dupsTested: 0,
        dupsDropped: 0
      }
    }
  },
  queryShapeHash: 'A2C6FEACD516B852393B090A140881943D95958319953C78E9D8F33DCC987D36',
  command: {
    find: 'employeeList_81',
    filter: {
      department: 'IT'
    },
    '$db': 'employeeDB_81'
  },
```

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Welcome | mongosh: localhost:27017

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```
    queryShapeHash: 'A2C6FEACD516B852393B090A140881943D95958319953C78E9D8F33DCC987D36',
    command: {
      find: 'employeeList_81',
      filter: {
        department: 'IT'
      },
      '$db': 'employeeDB_81'
    },
    serverInfo: {
      host: 'DESKTOP-05UTTVE',
      port: 27017,
      version: '8.0.10',
      gitVersion: '9d03076bb2d5147d5b6fe381c7118b0b0478b682'
    },
    serverParameters: {
      internalQueryFacetBufferSizeBytes: 104857600,
      internalQueryFacetMaxOutputDocSizeBytes: 104857600,
      internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
      internalDocumentSourceGroupMaxMemoryBytes: 104857600,
      internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
      internalQueryProhibitBlockingMergeOnMongoS: 0,
      internalQueryMaxAddToSetBytes: 104857600,
      internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
      internalQueryFrameworkControl: 'trySbeRestricted',
      internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
    },
    ok: 1
  }
employeeDB_81>
```